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Brain tumor segmentation with advanced nnU-Net: Pediatrics and adults tumors

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ABSTRACT

Automated brain tumor segmentation from magnetic resonance (MR) images plays a crucial role in precise diagnosis and treatment monitoring in brain tumor care. Leveraging the Brain Tumor Segmentation Challenge (BraTS) dataset, this paper introduces an extended version of the nnU-Net architecture for brain tumor segmentation, addressing both adult (Glioma) and pediatric tumors. Our methodology integrates innovative approaches to enhance segmentation accuracy. We incorporate residual blocks to capture complex spatial features, attention gates to emphasize informative regions and implement the Hausdorff distance (HD) loss for boundary-based segmentation refinement. The effectiveness of each enhancement is systematically evaluated through an ablation study using different configurations on the BraTS dataset. Results from our study highlight the significance of combining residual blocks, attention gates, and HD loss, achieving the best performance with a mean Dice and HD score of 83%, 3.8 and 71%, and 8.7 for Glioma and Pediatrics datasets, respectively. This advanced nnU-Net showcases the promising potential for accurate and robust brain tumor segmentation, offering valuable insights for enhanced clinical decision-making in pediatric brain tumor care.

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1. Introduction

Brain tumor segmentation from magnetic resonance (MR) images plays a crucial role in the field of brain tumor care as it enables clinicians to accurately locate, determine the extent of, and identify different types of tumors. This information is vital for initial diagnosis and treatment planning. However, the manual segmentation of brain tumors is a time-consuming and labor-intensive task that relies on the expertise of experienced neuro-radiologists. Challenges arise when dealing with large patient volumes, multi-parametric images with varying contrasts, and heterogeneous tumors. Moreover, inconsistencies in manual labeling between different raters and even within the same rater further complicates the interpretation and consensus on segmentation results. To overcome these challenges, automatic or computer-aided segmentation algorithms have emerged as potential solutions. These algorithms reduce the labor intensity and offer consistent results across cases. However, the development of such algorithms requires access to sufficiently annotated high-quality data to achieve satisfactory clinical performance.

In the realm of medical image segmentation, deep learning has emerged as a powerful tool, offering a wide range of benefits and applications. One of the primary advantages of deep learning in this field is its capacity to automatically extract relevant features from extensive medical image datasets, allowing it to isolate specific structures or regions of interest. The nnU-Net [1], introduced by Isensee et al., has gained popularity for brain tumor segmentation due to its self-configuring nature and adaptability to different datasets. Despite its success, the current nnU-Net architecture does have some limitations that can be addressed to further improve its performance.

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In this paper, we aim to extend the nnU-Net framework in three steps to overcome these limitations and improve brain tumor segmentation results. The first limitation we address is the use of a basic U-Net architecture in the baseline nnU-Net. Although U-Net has shown effectiveness in medical image segmentation tasks, it lacks the ability to capture complex spatial features effectively. To overcome this limitation, we leverage residual blocks, inspired by the ResNet architecture [2], to enhance the representation power of the nnU-Net. The addition of residual blocks allows the network to learn residual mappings, enabling the effective extraction of intricate spatial features critical for accurate tumor segmentation. The second limitation we address is the absence of an attention mechanism in the original nnU-Net. Attention mechanisms have proven effective in various computer vision tasks by allowing the network to focus on informative regions while suppressing irrelevant or noisy information [3]. We incorporate an attention mechanism into the nnU-Net architecture to improve the network's ability to selectively attend to tumor regions, enhancing the segmentation accuracy by giving more weight to relevant features. The third limitation we address is the use of the Dice coefficient [4] and cross-entropy loss in the nnU-Net's training process. While these loss functions have been widely used in medical image segmentation, they primarily focus on the overlap between predicted and ground truth masks, potentially neglecting fine details and boundary information [5,6] crucial for accurate tumor segmentation. To address this limitation, we add the Hausdorff distance (HD) loss [6] to the original loss function, which captures the distance between predicted and ground truth boundaries, encouraging the network to generate sharper and more precise tumor boundaries.

The contributions of this work are as follows:

1. Advanced nnU-Net architecture: We extend the baseline nnU-Net framework by incorporating three key enhancements: residual blocks, attention mechanism, and HD loss. These modifications address the limitations of the original nnU-Net and aim to improve the accuracy and robustness of brain tumor segmentation.

2. Evaluate the performance of the advanced nnU-Net on diverse datasets comprising both adult and pediatric brain tumors, compare it with state-of-the-art approach, and demonstrate the effectiveness of the proposed modifications.

2. Related work

Extensive exploration has been undertaken in the domain of brain tumor segmentation, focusing on various deep learning models for tumor segmentation. Deep neural networks have emerged as the preferred choice in recent years, as evidenced by their application in all recent works on the BraTS dataset, particularly benefiting from larger datasets. An ensemble of multiple models and architecture was proposed by [7], combining predictions from various 3D convolutional networks, including DeepMedic [8,9], FCN [10], and U-Net [11]. In [12], a 3D U-Net with an additional variational decoder branch was incorporated to introduce further supervision and regularization to the encoder branch. Additionally, [13] trained a two-stage cascaded U-Net, where the first stage generated a coarse segmentation mask, refined by the second stage. For their work, Isensee et al. [14] employed nnU-Net [1], a self-configuring framework automatically adapting U-Net to specific datasets, showcasing robust performance with minimal modifications to the conventional 3D U-Net, including BraTS-specific optimizations. [15] proposed an extended version of nnU-Net for brain tumor segmentation, modifying it with group normalization instead of batch normalization, an asymmetrically large encoder for the U-Net, and the inclusion of axial attention in the decoder.

The adaptability to new datasets and the presence of fully open-source codes and models made nnU-Net an ideal baseline for further experimentation. We introduced modifications to the baseline architecture by incorporating residual blocks, attention mechanisms, and HD loss. The effects of these modifications on the network's performance were evaluated through experimentation using cross-validation training data and unseen test data.

3. Method

In this section we propose the advanced nnU-Net and its components. First, we introduce the standard nnU-Net as the baseline, then three components including residual blocks, attention mechanism, and HD loss are explained in detail. Fig. 2 shows the components of the advanced nnU-Net.

3.1. Baseline: standard nnU-Net

Our approach makes use of nnU-Net, a fully automatic deep learning framework tailored for medical segmentation [1]. To establish a baseline for subsequent modifications, we initially apply nnU-Net without any adjustments. The design choices incorporated in nnU-Net are as follows: Brain voxels in each image are normalized through mean subtraction and standard deviation division, while non-brain voxels are assigned a value of 0. The network architecture follows a 3D U-Net structure, featuring an encoder and a decoder interconnected by skip connections. Feature extraction solely relies on plain convolutions, avoiding the utilization of recent architectural variations. Downsampling is achieved using strided convolutions, while upsampling is accomplished through convolution transposed. Deep supervision is implemented by introducing auxiliary segmentation outputs branching off at all resolutions, except for the two lowest levels in the decoder [14].

The input patch size is established as 128x128x128, and a batch size of 2 is employed. Leaky ReLUs [16] are chosen as nonlinearities, while instance normalization [17] is utilized to normalize feature maps.

In brain tumor segmentation, nnU-Net's region-based training allows for the merging of individual labels, offering benefits in detecting the whole tumor, tumor core, and enhancing tumor, rather than focusing solely on specific labels. This approach enables a shift in the importance of different labels during training, making it useful for tasks involving hierarchical classes, like segmenting organs and their substructures.

3.2. Residual blocks

In this work, the advanced nnU-Net architecture goes beyond the standard U-Net baseline by incorporating residual blocks inspired by the effective ResNet [2] architecture. Within a residual block, layers are structured so that the output of one layer is combined with a deeper layer before applying non-linearity, creating skip connections or shortcuts. These connections enable the network to learn residual mappings, capturing the differences between input and output of specific layers. By using residual connections, the network addresses the vanishing gradient problem in deep networks, allowing for effective modeling of complex spatial features. In brain tumor segmentation, the addition of residual blocks enhances the nnU-Net's ability to capture detailed spatial features crucial for accurate tumor segmentation. This inclusion enables the network to learn the residual information, improving its representational power and leading to more precise and robust tumor segmentation.

3.3. Attention

Attention mechanisms are integral to medical segmentation as they selectively focus on informative regions while suppressing irrelevant or noisy information. The main objective is to enhance the

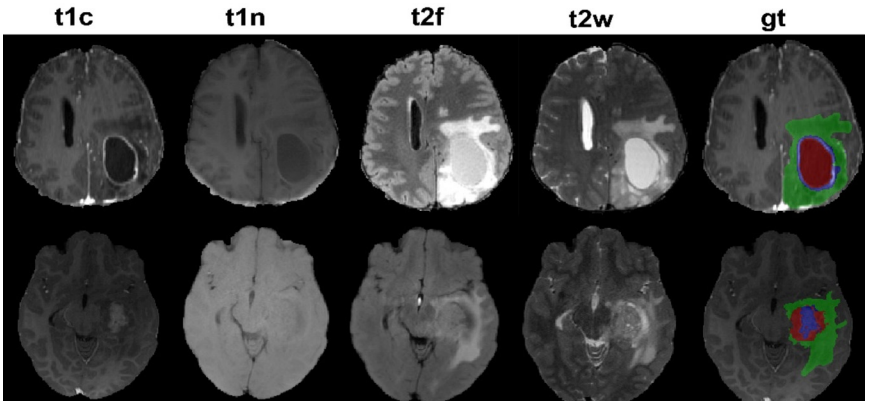


Fig. 1. Examples of BraTS dataset: (up): Adult Glioma, (down): Pediatric tumor.

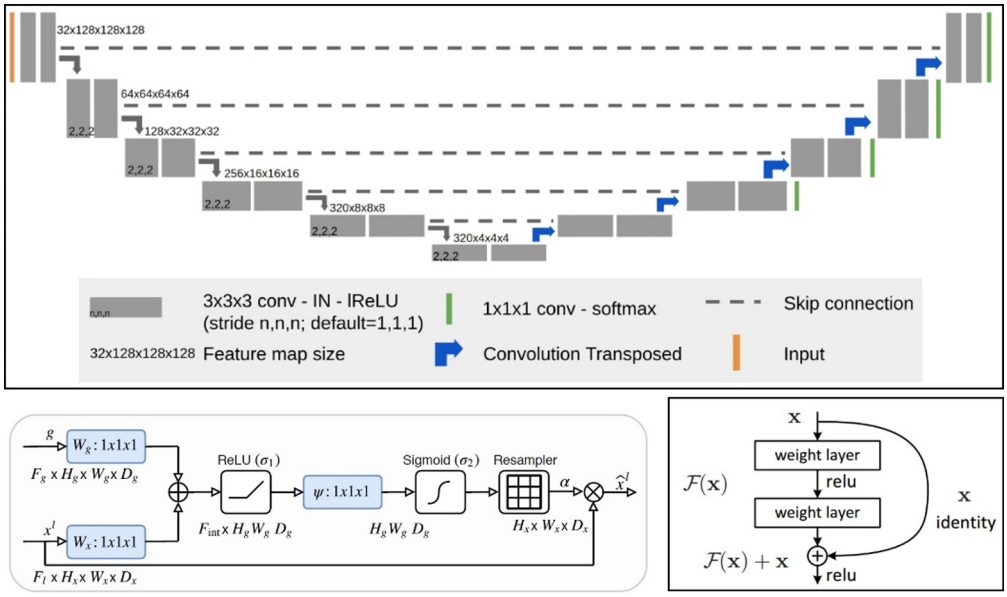


Fig. 2. Up: Baseline nnU-Net architecture. Down left: Attention gate. Down right: a Single residual block.

network's ability to prioritize essential features and regions for accurate segmentation. By assigning specific importance weights to different regions in an input image, attention mechanisms enable the network to allocate more resources and processing capacity to the most discriminative information. In brain tumor segmentation, attention mechanisms play a vital role in accurately identifying and segmenting tumor regions by concentrating on critical tumor features like shape, texture, and location. This targeted focus on informative tumor regions helps the network effectively distinguish between tumor and non-tumor areas, leading to significantly improved segmentation accuracy.

Attention mechanisms encompass two main approaches: soft attention and hard attention. Soft attention, a probabilistic method, utilizes soft shading to assign attention weights to different spatial regions or locations within an input image. On the other hand, hard attention, a deterministic approach, employs image cropping to focus solely on a single spatial location or region. In this study, we adopt the additive soft attention (AG) proposed by [18]. AG computes attention coefficients (alpha) by scaling the input features (xl) (Fig. 1). To select spatial regions, we analyze both the activations and contextual information provided by the gating signal (g), obtained from a coarser scale. Grid resampling of attention coefficients is achieved using trilinear interpolation.

Table 1
CE, Dice, and HD loss. Here $G = \{g_i\}$ and $S = \{s_i\}$ are ground truth and predicted segmentation, respectively. In HD loss, d_G and d_S are distance transforms of ground truth and segmentation.

Loss	Expression	Type
CE	$-\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} g_i^c \log s_i^c$	Distribution-based loss
Dice	$1 - \frac{2\sum_{i=1}^{N}\sum_{c=1}^{C} g_i^c s_i^c}{\sum_{i=1}^{N}\sum_{c=1}^{C} g_i^{c2} + \sum_{i=1}^{N}\sum_{c=1}^{C} s_i^{c2}}$	Region-based loss
HD	$\frac{1}{N}\sum_{i=1}^{N}[(s_i - g_i)\cdot(d_{Gi}^2 + d_{Si}^2)]$	Boundary-based loss

3.4. HD loss

In the domain of deep learning-based medical image segmentation, the selection of an appropriate architecture and weight tuning during training are critical factors. The training process involves minimizing a loss function that models the problem. While the baseline nnU-Net utilizes a hybrid loss comprising Cross-entropy (CE) as a distribution-based loss and Dice loss as a region-based loss, incorporating a boundary-based loss like the Hausdorff distance (HD) loss [6] brings several benefits, particularly in medical image segmentation tasks like brain tumor segmentation (Table 1).

The HD measure is a valuable performance metric in automated segmentation methods, indicating the largest segmentation errors. The HD loss estimates HD directly from the network output probability, encouraging precise tumor boundary localization and

improving segmentation quality. It captures fine details and works well in scenarios where boundary accuracy is crucial. The HD loss is less affected by spatial shifts and, when combined with CE and Dice, provides comprehensive segmentation evaluation, enhancing performance across different aspects.

3.5. Mathematical model of advanced nnU-Net

This section provides a detailed mathematical description of the proposed Advanced nnU-Net architecture for brain tumor segmentation. Our approach integrates three key components: residual blocks, attention gates, and the Hausdorff distance (HD) loss.

- Residual Blocks

The incorporation of residual blocks aims to capture complex spatial features within the neural network. A residual block can be mathematically represented as follows:

$$H(x) = F(x) + x$$

Where H(x) is the output of the residual block, F(x) represents the residual mapping to be learned, and x is the input to the block.

- Attention Gates

Attention gates are introduced to emphasize informative regions during the segmentation process. The attention mechanism can be mathematically described as follows:

$$Y = X.\sigma(W_g \times X + b_g)$$

Where X is the input feature map, σ denotes the sigmoid activation function, W_g and b_g are the learnable parameters, and Y represents the attention-weighted feature map.

- Hausdorff Distance (HD) Loss

The HD loss is employed to refine segmentation boundaries. The HD loss between two sets A and B is defined as:

$$HD(A, B) = \max\left(sup_{a\in A} inf_{b\in B} d(a, b), sup_{b\in B} inf_{a\in A} d(a, b)\right)$$

where d(a,b) represents the distance between points a and b.

- Overall

The advanced nnU-Net architecture coherently combines these components. The complete architecture can be represented as a composition of functions, each representing a layer in the neural network, with specific parameters and activation functions.

$$Output = FinalLayer\left(LayerN\left(\ldots Layer2\left(Layer1\left(Input\right)\right)\right)\right)$$

This mathematical model forms the foundation of our proposed approach, and its components are systematically integrated to enhance the segmentation performance.

4. Experiments

4.1. Data

We report our results on BraTS-Glioma [19–21] and BraTS-Pediatrics [22,23] datasets provided by Brain Tumor Segmentation Challenge 2023. Since currently the BraTS Challenge has only released the set of training cases, we use this data for training, validation, and testing.

Adult Glioma. MRI scans in the BraTS-Glioma comprise native (T1), post-contrast T1-weighted (T1Gd), T2-weighted (T2), and T2 Fluid Attenuated Inversion Recovery (T2-FLAIR) images. The evaluation primarily focuses on three sub-regions: the "enhancing tumor" (ET), the "tumor core" (TC), and the "whole tumor" (WT). The ET is characterized by areas with hyper-intensity in T1Gd compared to both T1 and "healthy" white matter in T1Gd. The TC represents the main tumor portion, given priority, and includes both the ET and the necrotic (NCR) regions of the tumor.

NCR is typically hypo-intense in T1-Gd compared to T1. The WT encompasses the complete extent of the disease, comprising the TC and the peritumoral edematous/invaded tissue (ED), which is typically depicted by a hyper-intense signal in FLAIR. In this work, we use 1100 cases for training and validation and 151 cases for testing the model.

Pediatrics. The dataset BraTS-PEDs 2023 comprises images from 228 pediatric high-grade glioma cases, encompassing pre- and post-gadolinium T1-weighted (referred to as T1 and T1CE), T2-weighted (T2), and T2-weighted fluid attenuated inversion recovery (T2-FLAIR). The data for these 228 patients were collected from CBTN, Boston's Children Hospital, and Yale University. From this pool of cases, we utilized 100 cases, with 80 cases designated for training and validation, while the remaining cases were reserved for testing.

During the evaluation, the focus rested on three regions: the "enhancing tumor" (ET), the "enhancing tumor/cystic component/necrosis" (TC), and the entire tumorous region known as the "whole tumor" (WT).

4.2. Evaluation metrics

We employed standard evaluation metrics to quantify the performance of the extended nnU-Net in brain tumor segmentation including Dice coefficient and the Hausdorff distance (HD). The Dice coefficient measures the overlap between predicted and ground truth tumor masks, while the HD measures the boundary localization accuracy between predicted and ground truth tumor boundaries.

4.3. Architecture & training

The tumor segmentation's baseline model consists of the 3D full resolution of nnU-Net, featuring an encoding and decoding path, where each path comprises 5 convolution blocks, each containing a 3*3*3 convolution layer. The training objective incorporates Dice and cross-entropy loss, focusing on the three class labels. For training, stochastic gradient descent is employed with an initial learning rate of 0.01 and a Nesterov momentum of 0.99. The training process spans 200 epochs, with each epoch comprising 100 iterations. Training patches are extracted by cropping from randomly selected training cases, and data augmentation techniques are consistently applied throughout the training procedure.

4.4. Results & discussion

In this section, we conduct a comprehensive analysis and comparison of the quantitative and qualitative results obtained from DeepMedic, the standard nnU-Net used as the baseline, and the advanced nnU-Net, including the ablation study.

Quantitative results. Table 2 presents the quantitative results of the advanced nnU-Net framework compared to the baseline nnU-Net and DeepMedic. The advanced nnU-Net consistently outperforms the baseline and DeepMedic in both evaluation metrics. The results demonstrate that the average segmentation of three regions provided by the advanced nnU-Net is 0.83, 3.8 for Dice and HD in BraTS-Glioma and 0.71, 8.7 in BraTS-Pediatric. The Dice coefficient is significantly improved, indicating better overall segmentation accuracy. A higher Dice value indicates a better agreement between the predicted tumor segmentation and the ground truth mask.

The superior Dice value signifies that the advanced nnU-Net provides more reliable and consistent segmentation results compared to the baseline and DeepMedic. In the other side, the HD

Table 2
Segmentation results of DeepMedic, baseline nnU-Net (*), and advanced nnU-Net(**) on BraTS Glioma and Pediatric datasets: "WT" stands for whole tumorous area, "ET" stands for enhancing tumor, and "NEC" stands for non-enhancing components.

Dataset	Method	Average		WT		ET		NEC (TC)	
		DSC	HD95	DSC	HD95	DSC	HD95	DSC	HD95
BraTS-GLI	DeepMedic	0.74	4.8	0.79	4.4	0.69	5.2	0.75	4.8
	nnU-Net*	0.76	4.4	0.81	3.5	0.72	4.2	0.77	5.5
	nnU-Net**	**0.83**	**3.8**	0.90	3.1	0.79	4.4	0.81	3.9
BraTS-PED	DeepMedic	0.63	8.8	0.75	8.2	0.54	8.9	0.61	9.2
	nnU-Net*	0.64	9.03	0.79	8.5	0.52	9.2	0.61	9.4
	nnU-Net**	**0.71**	**8.7**	0.85	8.5	0.60	8.9	0.69	8.7

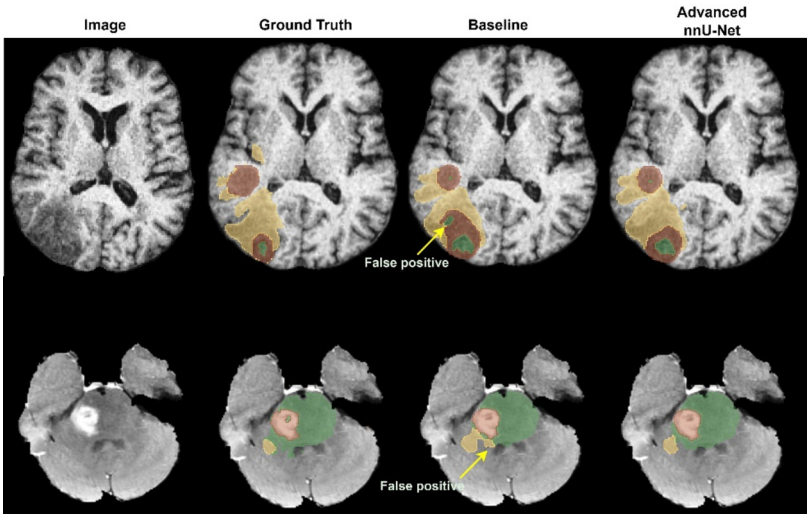


Fig. 3. Segmentation result provided by standard nnU-Net and advanced nnU-Net. While we ran the model on 5-fold cross validation, this result was obtained from the first fold. Up: BraTS-Glioma, Down: BraTS-Pediatric.

value measures the dissimilarity between the predicted boundary of the segmented tumor and the ground truth boundary. The lower HD value provided by the advanced nnU-Net means that it achieves superior boundary localization accuracy compared to the baseline nnU-Net and DeepMedic approaches. The combination of achieving better Dice and HD values demonstrates that the advanced nnU-Net not only improves the overall segmentation accuracy but also produces sharper and more precise tumor boundaries. This enhanced segmentation performance is crucial for accurate brain tumor diagnosis and treatment planning. To evaluate the statistical analysis of the results in Table 2, and to ascertain the significance of disparities among DeepMedic, basic nnU-Net, and advanced nnU-Net, we conducted a Kruskal-Wallis test. This non-parametric test is particularly valuable when comparing two or more independent samples of equal or varying sizes. A significant outcome from the Kruskal-Wallis test implies that, at the very least, one sample stochastically dominates another. The results of the analysis unveiled statistically significant differences in mean Dice (DSC) and HD95 across various methods in the context of BraTS-Glioma, with a noteworthy p-value of 4.8E-3 for Dice and 2.9E-2 for HD95. This statistical significance underscores the robustness of our findings and substantiates the superiority of the advanced nnU-Net in achieving better segmentation results in the BraTS-Glioma dataset. However, in the case of BraTS-Pediatric, while the statistical analysis revealed that there is a significant difference between Dice with p-value of 3.6E-2, no significant difference appears in HD95 value among the methods.

Qualitative results. Fig. 3 shows qualitative results and comparison of the baseline and advanced nnU-Net segmentation on representative cases from the dataset. The segmentation outputs demonstrate more precise tumor segmentation with lower number of false positives by Advanced nnU-Net, highlighting the effectiveness of the contributions including the attention gate and HD loss.

Execution time analysis. Given the intricate architecture of the advanced nnU-Net in comparison to the simpler structures of the basic nnU-Net and DeepMedic framework, a comprehensive analysis of their execution times was conducted. The simulations were meticulously performed under identical parameters, iterations, and cycles. Upon evaluating the average execution running times of the models across both Pediatric and Adults brain tumor datasets, it was observed that DeepMedic exhibited the shortest running time, completing each cycle in approximately 27 seconds. In contrast, the basic nnU-Net and advanced nnU-Net required more time to perform the same task, taking 84 and 119 seconds, respectively. This comparison sheds light on the computational efficiency of these models, with DeepMedic showcasing the swiftest execution.

Ablation study. The ablation study evaluates the contribution of each component of the proposed method by systematically removing them. Specifically, we examined the effect of removing residual blocks, attention gates, and HD loss. Tables 3–4 indicate the effect of each component on the overall segmentation for two datasets.

In BraTS-Pediatric, the baseline nnU-Net configuration demonstrated moderate segmentation performance, with a mean Dice (DSC) value of 0.64 $\pm$ 0.37 and a mean HD of 9.03 $\pm$ 12.29. These results indicate that the nnU-Net without any specific enhancements can provide initial segmentation results, but there is room for improvement. When the residual blocks were introduced into the nnU-Net architecture, we observed a slight increase in the mean Dice value to 0.65 $\pm$ 0.35 and a slight reduction in the mean HD value to 9.00 $\pm$ 12.05. Although the improvement was not substantial, it suggests that residual blocks contributed to capturing complex spatial features and enhancing segmentation performance. The addition of attention gates to the nnU-Net re-

Table 3
Segmentation results (average region) by ablation study of the advanced nnU-Net on BraTS-Glioma: "RES" refers to residual block, "ATTN" as attention block, and "HD" loss.

Baseline	Module			Metrics (Mean ± Std)	
	RES	ATTN	HD	DSC	HD95
nnU-Net	-	-	-	0.76 ± 0.18	4.42 ± 5.25
nnU-Net	+	-	-	0.77 ± 0.15	4.40 ± 5.64
nnU-Net	-	+	-	0.79 ± 0.22	4.01 ± 5.40
nnU-Net	-	-	+	0.79 ± 0.20	4.35 ± 5.05
nnU-Net	+	+	-	0.80 ± 0.17	4.00 ± 6.27
nnU-Net	+	-	+	0.79 ± 0.20	3.89 ± 6.01
nnU-Net	-	+	+	0.82 ± 0.22	3.82 ± 5.84
nnU-Net	+	+	+	0.83 ± 0.20	3.80 ± 5.85

Table 4
Segmentation results (average region) by ablation study of the advanced nnU-Net on BraTS-Glioma: "RES" refers to residual block, "ATTN" as attention block, and "HD" loss.

Baseline	Module			Metrics (average)	
	RES	ATTN	HD	DSC	HD95
nnU-Net	-	-	-	0.64 ± 0.37	9.03±12.29
nnU-Net	+	-	-	0.65 ± 0.35	9.00±12.05
nnU-Net	-	+	-	0.67 ± 0.30	8.92±12.01
nnU-Net	-	-	+	0.67 ± 0.34	8.88±11.89
nnU-Net	+	+	-	0.67 ± 0.30	8.79±.11.64
nnU-Net	+	-	+	0.68 ± 0.29	8.77±11.50
nnU-Net	-	+	+	0.70 ± 0.24	8.72±11.34
nnU-Net	+	+	+	0.71 ± 0.28	8.70±11.27

Table 5
Segmentation comparison of our method with state-of-the-works. "WT" stands for whole tumorous area, "ET" stands for enhancing tumor, and "NEC" stands for non-enhancing components.

Dataset	Method	WT		ET		NEC (TC)	
		DSC	HD95	DSC	HD95	DSC	HD95
BraTS-GLI	FCN8	0.66	8.1	0.59	10.3	0.63	9.5
	U-Net	0.70	7.5	0.63	8.6	0.69	7.2
	Dense U-Net	0.73	7.0	0.66	6.9	0.71	7.2
	SCU-Net [24]	0.79	5.1	0.71	4.1	0.77	5.4
	Basic nnU-Net	0.81	3.5	0.72	4.2	0.77	5.5
	Our method	**0.90**	**3.1**	**0.79**	**4.4**	**0.81**	**3.9**
BraTS-PED	FCN8	0.68	8.6	0.44	12.4	0.49	13.7
	U-Net	0.69	8.8	0.41	13.7	0.48	13.5
	Dense U-Net	0.70	8.2	0.43	13.0	0.51	13.1
	SCU-Net [24]	0.77	8.1	0.53	8.7	0.58	12.6
	Basic nnU-Net	0.79	8.5	0.52	9.2	0.61	9.4
	Our method	**0.85**	**8.5**	**0.60**	**8.9**	**0.69**	**8.7**

sulted in further improvement in segmentation performance. The mean Dice value increased to 0.67 ± 0.30, and the mean HD value decreased to 8.92 ± 12.01. Incorporating the Hausdorff distance loss into the nnU-Net architecture resulted in similar segmentation performance as the attention gates. The mean Dice value remained at 0.67 ± 0.34, while the mean HD value decreased to 8.88 ± 11.89. HD loss emphasizes precise boundary localization, which contributes to better segmentation results. The best segmentation performance was achieved when all three components (RES, ATTN, and HD) were integrated into the nnU-Net architecture. The model achieved a mean Dice value of 0.71 ± 0.28 and a mean HD value of 8.70 ± 11.27. This demonstrates the synergy between the components, each enhancing specific aspects of the segmentation process.

In BraTS-Glioma dataset, the addition of residual blocks to the nnU-Net architecture provided a slight improvement in segmentation performance. The enhanced ability of residual blocks to capture complex spatial features likely contributed to the improved segmentation accuracy and boundary localization. However, incorporating attention gates had a more substantial positive impact on the model's performance. The attention mechanism enabled the network to focus on informative regions while suppressing noise, leading to better tumor segmentation results. This improvement highlights the importance of attention mechanisms in medical image segmentation. The utilization of the HD loss in the advanced nnU-Net further improved segmentation accuracy and boundary delineation. The boundary-based loss function encouraged the network to produce more precise boundary predictions, resulting in sharper and more accurate tumor boundaries. Overall, the ablation study highlights the attention gate and HD loss play crucial roles in enhancing segmentation accuracy and boundary delineation.

Comparison with other methods. We conducted a comprehensive comparison of segmentation results obtained from our advanced nnU-Net with state-of-the-art models, encompassing FCN8, U-Net, Dense U-Net, SCU-Net, and basic nnU-Net, across both pediatric and adult tumor datasets. In both datasets, Table 5 indicates that our findings consistently demonstrate that the advanced nnU-Net surpasses other methods, exhibiting superior performance in both Dice and HD95 metrics. The results unequivocally support the efficacy of our approach in achieving the best overall segmentation performance.

5. Conclusion

This paper presents an advanced nnU-Net framework for automated brain tumor segmentation, addressing the limitations of the baseline architecture. The extension involves the incorporation of residual blocks, attention gates, and the Hausdorff distance (HD) loss. By leveraging residual blocks, the proposed nnU-Net architecture captures complex spatial features more effectively, enhancing the representation power and enabling more accurate tumor segmentation. The inclusion of attention gates allows the network to selectively attend to informative tumor regions, improving segmentation accuracy by emphasizing relevant features. Moreover, the integration of the HD loss enhances boundary localization and detail, resulting in sharper and more precise tumor boundaries. Our experimental results have shown satisfactory performance of the advanced nnU-Net framework in automated brain tumor segmentation. The segmentation results demonstrate improved accuracy and robustness compared to the baseline nnU-Net and state-of-the-art approaches.

Human and animal rights

The authors declare that the work described has not involved experimentation on humans or animals.

Informed consent and patient details

The authors declare that this report does not contain any personal information that could lead to the identification of the patient(s) and/or volunteers.

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Author contributions

All authors attest that they meet the current International Committee of Medical Journal Editors (ICMJE) criteria for Authorship.

Declaration of competing interest

The authors declare that they have no known competing financial or personal relationships that could be viewed as influencing the work reported in this paper.

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